

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DIGITAL SIGNAL PROCESSING

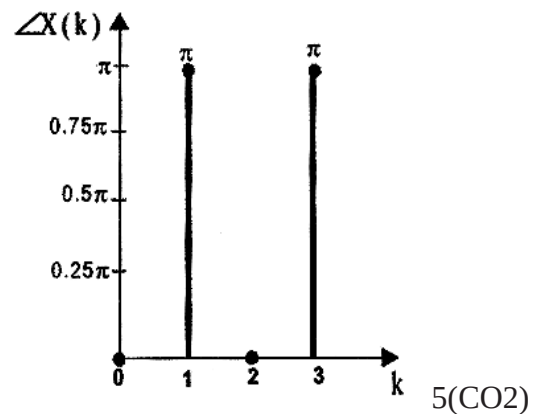
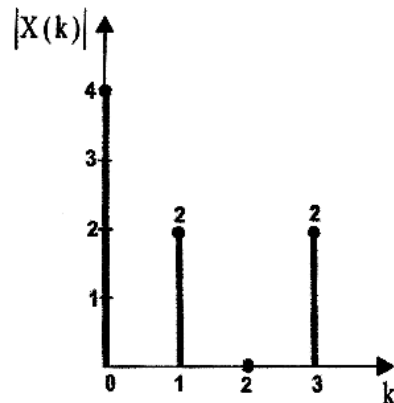
Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Assume suitable data wherever necessary.
- (3) Illustrate your answers with neat sketches.
- (4) Use of Non – programmable calculator is permitted.

1. (a) Compute circular convolution of $h(n) = \{-1, -1\}$ with $x(n) = \{-3, 2, -1\}$ using FFT-IFFT algorithm. 5(CO1,2)
- (b) Reconstruct and plot the causal signal $x(n)$ from its DFT shown below,



2. (a) Construct the Direct Form-I structure for the digital system given by –

$$H(z) = \frac{2z^3 - 4z^2 + 11z - 8}{(z - 8)(z^2 - z + 3)}$$

Optimize this realization using Direct form - II.

6(CO5)

- (b) Realize the optimum digital system structure for the system given by –

$$H(z) = \frac{1}{3} + \frac{1}{4}z^{-1} + \frac{3}{2}z^{-2} + \frac{3}{2}z^{-3} + \frac{1}{4}z^{-4} + \frac{1}{3}z^{-5} \quad 4(\text{CO5})$$

3. (a) Design a digital Butterworth IIR low pass filter applying impulse invariant transformation by taking $T = 1$ second, to satisfy the following specifications :

$$0.707 \leq |H(e^{j\omega})| \leq 1.0 \quad \text{for } 0 \leq |\omega| \leq 0.3\pi$$

$$|H(e^{j\omega})| \leq 0.2 \quad \text{for } 0.75\pi \leq |\omega| \leq \pi$$

Realize direct form-II structure of the filter and sketch the nature of its magnitude response. 8(CO1,3)

- (b) Compare Butterworth filter and Chebyshev filter. 2(CO1,3)

4. Design a FIR bandstop filter to reject frequencies in the range 1.5 kHz to 3 kHz and sampling frequency of 8 kHz with 7 samples using Fourier series method. Realize the filter structure, determine the frequency response and sketching the magnitude response. 10(CO1,3)

5. (a) A single $x(n)$ having its magnitude spectrum shown in Fig. 5 (a) is passed through a downsampler having factor of 2. Sketch the spectrum of the signal at the output of this downsampler. Comment on the output spectrum if the downsampling factor is changed to 3.

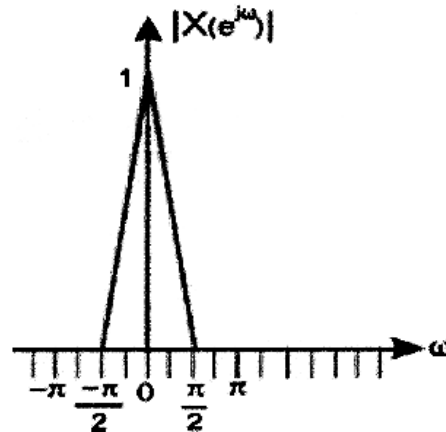


Fig. 5 (a)

4(CO4)

- (b) Describe Quadrature Mirror Filter Bank as an application of multirate system. Express the relation of distortion transfer function and aliasing component. 6(CO4)
6. (a) Explain VLIW architecture of DSPs along with function diagram. 5(CO1)
- (b) Describe Braun multiplier with an example of 5 bit number. 5(CO1)

